

Capstone Project Report

Title: Internet Adoption Trends – World Bank WDI Analysis

1. Objective

The purpose of this project is to analyze global internet usage trends using World Bank's World Development Indicators (WDI) dataset. The dashboard highlights growth patterns over time and compares adoption rates across countries, with a focus on 2024.

2. Dataset

- **Source:** World Bank World Development Indicators (WDI)
- **File:** `wdi_cleaned.csv`
- **Key Fields:** Country Name, Year, Indicator, Value

3. Data Cleaning

- Removed duplicates and irrelevant indicators.
- Standardized year values.
- Filtered dataset to focus on **Internet Users (% of population)**.

4. Dashboard Components

- **KPI Card:** Displays average internet usage (%).
- **Line Chart:** Shows internet adoption trend from 1996–2024.
- **Bar Chart:** Compares internet usage across countries in 2024.
- **Slicers:** Year and Country filters for interactive exploration.

5. Key Insights

- Internet usage has grown significantly since the early 2000s.
- Global average in 2024 is **71.2%**, but adoption varies widely.
- Developed regions show near-universal access, while lower-income countries lag behind.
- Africa and South Asia remain below the global average, highlighting digital inequality.

6. Recommendations

- **Infrastructure Investment:** Expand broadband and mobile networks in underserved regions.

- **Digital Literacy Programs:** Empower communities with skills to use online resources.
- **Affordable Access:** Policies to reduce costs of internet services and devices.
- **Regional Collaboration:** Share best practices across countries to accelerate adoption.

7. Conclusion

Internet adoption is a critical driver of economic growth, education, and innovation. While global progress is evident, disparities remain. Targeted interventions can close the digital divide and ensure inclusive access to the benefits of connectivity.